PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: SyncBite-AI

Version: 1.0

1. Project Overview

- **Problem Statement:**
  In university towns across Malaysia, SMEs such as dessert shops, food trucks and small pop-up stalls are facing a critical challenge: extreme foot traffic volatility driven by severe information asymmetry. Their daily survival depends entirely on campus dynamics. Often, merchants spend heavy upfront costs preparing perishable goods, only to arrive and discover a sudden campus holiday, a shifted exam week, or afternoon classes moving online due to heavy thunderstorms. Because they rely entirely on manual, "gut-feeling" forecasting rather than real-time data, this disconnect leads to massive perishable food waste, unrecoverable sunk costs, and lost profit margins
- **Target Domain:** AI for Economic Empowerment & Decision Intelligence.
- **Proposed Solution Summary:**
  SyncBite AI is a Decision Intelligence platform designed to completely eliminate the campus-SME information gap. It acts as a real-time strategic bridge by providing:
  - **Structured Campus Intel:** Aggregating academic calendars, exam weeks, and scheduled campus activities (e.g. club recruitment days).
  - **Live Emergency Alerts:** Monitoring sudden anomalies like extreme weather or unexpected online-class shifts.

  By syncing this real-time campus context with the merchant's current inventory, the platform translates volatile signals into actionable, AI-driven business pivots. This empowers SMEs to instantly adjust pricing, create targeted delivery bundles, minimize inventory waste, and protect their daily profitability.

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2. Background & Business Objective

- **Background of the Problem:**
  Running a small food business near a university campus has never been easy. For years, campus-adjacent merchants have planned their day the same way — based on experience and instinct. It worked when campus life was predictable. But today, exams get rescheduled, semester breaks shift, and a single thunderstorm can move an entire afternoon of students online. The rhythm these merchants depended on no longer exists, and they are left making critical daily business decisions with no real-time information to guide them.
- **Importance of Solving This Issue:**
  Every unsold tray of food is a loss these merchants cannot recover. Unlike larger businesses, campus SMEs have no financial buffer, their daily income depends entirely on getting their preparation right. Giving them access to real-time campus intelligence means less waste, more predictable revenue, and the confidence to run their business without constantly absorbing avoidable losses.
- **Strategic Fit / Impact:**
  SyncBite AI supports Malaysia's push for SME digital empowerment by closing the information gap between campus operations and merchant planning. Powered by Z.AI's GLM, the platform is built to scale across university towns nationwide, turning real-time campus signals into smarter, data-driven decisions for the businesses that need it most.

3. Product Purpose

3.1. Main Goal of the System:

To act as a real-time Decision Intelligence partner for campus-adjacent SMEs, eliminating guesswork from daily business planning by translating live campus activity into clear, actionable inventory and pricing decisions.

3.2. Intended Users (Target Audience):

- Food truck operators running near or within university campuses
- Campus canteen operators
- Dessert shops & beverage stalls around campuses
- Pop-up stall vendors whose customer flow is tied to student activity
- Any campus-adjacent or campus-dependent SME

4. System Functionalities

4.1. Description: SyncBite-AI operates as a real-time decision engine that continuously monitors campus activity and cross-references it with the merchant's current inventory. The system interprets live contextual signals, such as academic schedules, weather conditions, and sudden campus announcements, and translates them into specific, immediately actionable business recommendations. Every suggestion the platform makes is accompanied by a clear explanation, so merchants always understand the reason behind each decision.

4.2. Key Functionalities:

- **Smart Risk Assessment & dashboard:** Provides merchants with a real-time daily overview displaying live order count, waste risk level, and average ticket value. Urgent signals such as weather warnings, expiring stock, and upcoming campus events are surfaced as quick alert cards for immediate action.
- **Campus Intelligence & Academic Calendar:** Syncs with the merchant's selected university to display a live academic calendar. Each day is colour-coded by campus condition, giving merchants advance visibility into upcoming demand shifts.
- **AI Assistant:** A multimodal conversational assistant that accepts text, voice, photos, and file inputs. Merchants can request promo ideas, ask business questions, or upload stock photos for instant AI-driven guidance.
- **AI Strategy Engine:** Produces a daily business pivot recommendation based on the merchant's live inventory and current campus context. Every strategy includes a specific action, the GLM's reasoning, a secondary tip, and a predicted financial recovery value.

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4.3. AI Model & Prompt Design

4.3.1. Model Selection: SyncBite-AI is powered by Z.AI's General Language Model (GLM). The GLM was selected for its strong contextual reasoning capabilities, allowing the system to process and connect multiple unstructured inputs simultaneously, such as campus announcements, weather reports, and inventory data, and produce coherent, business-specific recommendations in real time.

4.3.2. **Prompting Strategy:** SyncBite-AI adopts a zero-shot prompting strategy. Rather than relying on pre-loaded examples, the GLM is given a structured context window containing the merchant's current inventory state, live campus signals, and a clearly defined output format. This approach was chosen because campus conditions change rapidly and unpredictably. A few-shot approach with fixed examples would struggle to cover the full range of real-world scenarios. Zero-shot prompting allows the GLM to reason flexibly across novel situations without being anchored to prior examples, making it the right fit for a platform built around real-time, context-sensitive decision making.

4.3.3. **Context & Input Handling**: The system accepts both structured inputs (inventory levels, product shelf-life, cost price, historical sales data) and unstructured inputs (campus news, weather reports, social media sentiment, emergency announcements). All inputs are aggregated and summarized into a concise context package before being passed to the GLM, keeping the prompt within a manageable token window. Inputs that exceed the defined limit are automatically chunked and prioritized by recency and relevance, ensuring the most critical signals always reach the model first.

4.3.4. **Fallback & Failure Behavior:** SyncBite AI handles API instability, off-topic generations, and model hallucinations through a three-layer failsafe design.

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- **Validation & Auto-Retry Protocol:** The GLM is explicitly prompted to return outputs in a strict, predefined structure (Action, Reasoning, Impact). If the backend receives an unparseable response or encounters a server timeout (e.g., 504 error), the system initiates an automatic retry loop (up to 3 attempts) with an escalating time delay to allow the server to recover.
- **Graceful Degradation (Mock Fallback):** If the Z.AI API remains completely unresponsive or repeatedly fails validation after the maximum retries, the system degrades gracefully. Instead of crashing or showing a blank screen, the Python backend triggers a hardcoded fallback function (get_mock_advice()). This provides the user with a safe, baseline mitigation strategy (e.g., a standard 15% perishable goods discount), ensuring continuous operational support.
- **Human-in-the-Loop Escalation:** To mitigate the risk of financial damage from subtle hallucinations (e.g., the AI suggesting a discount that drops below the cost price), SyncBite-AI is strictly designed as an advisory tool. The system does not possess autonomous execution capabilities over the merchant's POS or delivery platforms. The SME owner acts as the final escalation checkpoint, reviewing the AI's reasoning and predicted impact before manually clicking "Execute Campaign" to validate the decision.

5. User Stories & Use Cases

- **User Stories:** *Example:* As a campus dessert stall owner, I want to receive a real-time AI-generated strategy based on my current stock and today's campus conditions, so that I can immediately act to clear at-risk inventory and protect my daily revenue, without spending time manually analysing the situation myself."
- **Use Cases (Main Interactions):**

  Inventory Sync Flow: Merchant logs current stock items with quantity, expiry status, and pricing. The system classifies each item by risk tier and the merchant taps "Sync with Z.AI Brain" to pass the live inventory state to the GLM for processing.

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Campus Intelligence Flow: Merchant selects their linked university and views a live colour-coded academic calendar. Upcoming exam weeks, holidays, and crisis days are visible in advance, allowing the merchant to adjust stock preparation before the shift in demand arrives.

AI Strategy Generation Flow: The GLM cross-references the merchant's synced inventory with live campus context and generates a Today's Strategic Pivot: a specific action, clear reasoning, secondary tip, and predicted financial recovery value.

- **Use Case Diagram:** *(Optional but encouraged for participants to visualize the interaction)*.

6. Features Included (Scope Definition)

- **Real-Time Daily Snapshot:** A dashboard displaying live order count, waste risk level, and average ticket value to monitor business health.
- **Quick Alert System:** Actionable cards surfacing urgent signals such as weather warnings, expiring stock, and upcoming campus events.
- **Smart Foot Traffic Forecasting:** A 5-day trend visualization combined with AI-powered demand prediction to anticipate customer flow.
- **Automated Risk Inventory:** Manual stock input paired with automated risk tier classification (High, Medium, Safe) based on item shelf life.
- **Campus Intelligence Hub:** A dedicated tab featuring university selection and a live, color-coded academic calendar to track local demand drivers.
- **Multimodal AI Assistant:** A conversational interface supporting text, voice, photo, and file inputs for on-demand, contextual business guidance.

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- **Z.AI Strategy Engine:** An intelligence module generating daily business pivot recommendations complete with the action, reasoning, secondary tips, and predicted financial impact.
- **Human-in-the-Loop Execution:** A strict operational model where merchants must manually confirm and execute all AI recommendations before any market action is taken.

7. Features Not Included (Scope Control)

- No direct integration with POS systems, delivery platforms, or payment gateways.
- No automated execution of promotions, all AI recommendations require manual merchant approval before action is taken.
- No multi-user or team account support in the current version.
- No real-time social media sentiment scraping or live student forum monitoring.
- No support for non-campus or non-perishable inventory business types.
- No predictive reordering or automatic supplier communication features.
- Academic calendar data is manually synced and does not pull live updates directly from university systems.
- No native mobile applications (iOS/Android), the system is a responsive web application and will not be compiled or deployed to app stores for the preliminary round.
- No autonomous model fine-tuning, the Z.AI engine relies on dynamic prompt engineering rather than continuous training on the merchant's private historical data.
- No IoT hardware integration, the system does not utilize smart scales or connected refrigerators to automatically detect inventory depletion or spoilage.

8. Assumptions & Constraints

8.1. System Assumptions:

Stable Network Connectivity: The system assumes the merchant has a stable internet connection on their mobile device or terminal to send and receive real-time updates from the Vercel frontend to the Render backend.
Accurate Data Entry: The financial predictions assume the merchant accurately inputs the correct initial cost price and target selling price for their inventory items.

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Consistent User Behavior: The AI models assume that student purchasing behavior remains relatively consistent with the provided campus context (e.g., assuming students will actually stay indoors and order delivery during a thunderstorm).

8.2. Technical Constraints:

- **LLM Token Limits (Cost Control):** To manage API costs and prevent context-window overflow, the frontend enforces a strict data-filtering mechanism. Only inventory items categorized as "Medium Risk" or "High Risk" are packaged into the JSON payload sent to the Z.AI GLM. "Safe" items are deliberately excluded to save tokens.
- **Inference Latency:** The Z.AI GLM (ilmu-glm-5.1) requires processing time. The system is constrained by a 2-to-5 second latency window during strategy generation, requiring the UI to utilize optimistic loading animations to prevent user abandonment.
- **Stateless Backend:** The Python FastAPI backend deployed on Render operates statelessly for this MVP. It does not store historical conversation logs or inventory states across sessions; all state is currently managed locally within the React frontend's memory.

8.3. Business Constraints:

- **Advisory Only (No Execution Autonomy):** For liability and safety reasons, SyncBite-AI is strictly constrained to an advisory role. It cannot autonomously alter prices on third-party delivery platforms or automatically deduct money from a supplier account.
- **Campus API Availability:** The "Campus Intelligence Sync" relies on the assumption that external university calendars and local weather alerts remain publicly accessible or can be scraped reliably. If these external sources change their data structures, the context engine may experience temporary failures.

9. Risks & Questions Throughout Development

- **Hallucination Risk:** How do we ensure the GLM does not recommend a flash sale price that falls below the merchant's actual cost price, resulting in a guaranteed loss rather than cost recovery?
- **Data Quality Dependency:** What happens when a merchant inputs inaccurate stock quantities or wrong cost prices, does the AI still generate a financially sound recommendation, or does garbage-in produce garbage-out advice?

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- **Campus Data Reliability:** If a university suddenly changes its academic calendar or moves classes online without a public announcement, how does the system detect and respond to that unscheduled disruption?
- **Merchant Adoption Friction:** Will time-pressured SME owners, many of whom are not tech-savvy, consistently log their inventory every morning, and if they skip this step, does the platform still provide meaningful value?
- **Weather API Dependency:** If the external weather data source experiences downtime or returns inaccurate forecasts, how does this affect the reliability of the foot traffic prediction and the strategy generated?
- **Scope Creep Risk:** As the platform proves useful, merchants may request features beyond advisory recommendations, such as automated posting to WhatsApp groups or direct delivery platform integration, which could compromise the current stateless, advisory-only architecture.

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